



BIOBRIDGE OS

Closing the wet-lab-to-dry-lab data handoff gap

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What we'll cover

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The Problem

Why the wet-lab → dry-lab
handoff breaks down today

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The Solution

BioBridge OS, demonstrated
live

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Validation & Market

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Roadmap & Ask

Where this goes next, and
how you can help

THE PROBLEM

A structural gap, not a personal failure

Wet-lab scientists generate messy, inconsistent tabular data: plate layouts, omics readouts, screening exports.

Computational biologists spend the majority of their time cleaning that data before any real analysis begins.



Bench data, captured by hand



Pipelines, blocked on cleanup

60%

of computational time spent cleaning data, not analyzing

Neither persona is at fault since no instrument knows experimental design, and no spreadsheet has a schema.

PERSONA 1

Dr. Elena Vance

Wet-lab biologist



- Manually adds metadata her instrument can't capture — treatment, replicate #, batch.
- No way to audit her own data before sending it — “clean” is undefined until it reaches Marcus.
- Custom note cells get silently dropped downstream, with no record of when or why.
- Multi-day/multi-batch runs mean manually merging several exports into one file by hand.

PERSONA 2

Marcus Chen

Computational biologist



- Inconsistent naming everywhere — Control_Rep1, ctrl_r2, CONTROL_REP1, vehicle, C1.
- Non-numeric values land in numeric columns — “LOW_QUALITY,” “ERROR,” “N/A,” “<LOD.”
- No schema or data dictionary — is this column raw counts, or a normalized ratio?
- No audit trail — every data question becomes a Slack thread and a guess.

What this gap actually costs



50–60%

of computational time spent cleaning,
not analyzing



2–4 days

round-trip turnaround for a single
data-quality question



0

audit trail in most observed workflows
today

THE SOLUTION

BioBridge OS

A shared workspace where both personas define and resolve data rules together in minutes, not days.



No regex, by default

Messy data is resolved with cluster cards and click-to-merge suggestions and never enforce pattern syntax, for the default user.



Full audit trail

Every change is attributed, timestamped, and reasoned which removes ambiguity and blame from the handoff.



Real, exportable code

Every visual action has a live, reproducible Python/Pandas counterpart to validate the system's actions via code if necessary.

Automated fixes & suggestions

The moment a file loads, BioBridge OS surfaces every issue in plain language — before Elena ever sends it.

Pre-flight check — 14 issues before you send
Fix these now so Marcus doesn't have to Slack you back in three days. Plain-language summary of what we found:

- 11x Not in protocol
- 1x Hidden spaces
- 1x Empty cell
- 1x Missing replicate

Notes column "Notes" will be preserved in export — your bench judgments won't be dropped.
Multi-batch file detected (Batch) — watch for naming drift across days.
Checking against protocol: 96-well viability screen v1

2 fixes can be applied automatically — see Suggested fixes below.

Suggested fixes
12 recommendations — 2 can be applied automatically Apply all safe fixes (2) Dismiss

△ Notes column "Notes" detected — will be preserved in all exports.

Multi-batch naming drift detected
Different lab members labeled the control differently across batches (Control, CONTROL_REP1). Recommend standardizing to "Control". Suggested: **Control**

Batch consolidation: use "Control" high confidence Auto-fix OK Apply
Different lab members labeled the control differently across batches (Control, CONTROL_REP1). Recommend standardizing to "Control".
Across batches: Run_A: Control/ CONTROL_REP1 - Run_B: Control

Remove hidden spaces high confidence Auto-fix OK Apply
Trim invisible spaces in 1 cell(s) — one-click fix.

Standardize "Drug X" medium confidence Apply
Pick the closest protocol label for "Drug X".
Use "Control" Use "Drug A" Use "Drug B" Use "Drug C"

Standardize "typed fast at bench" medium confidence Apply

Why this matters

- Detection runs automatically so no manual scanning required.
- Issues are explained in bench language, not data-engineering jargon.
- High-confidence fixes can be auto-applied; ambiguous ones are routed to a human judgment call.

An attributable audit trail

Every resolution — human or automatic — is logged: what changed, who decided it, and why.

Audit Trail		Export JSON	Export CSV	
Type	Target	Before → After	Actor	Time (s)
merge_cluster	Row 1, Treatment_Group	CONTROL_REP1 → Control	Dr. Elena Vance	0.0
merge_cluster	Row 4, Treatment_Group	CONTROL_REP1 → —	Dr. Elena Vance	0.0
promote_to_protocol	CONTROL_REP1	CONTROL_REP1 → Control	Dr. Elena Vance	2.0

Why this matters

- Removes blame from the handoff so decisions are visible, not guessed at.
- Exportable as CSV/JSON alongside the cleaned dataset.
- Entries are immutable so undo creates a new entry, history is never erased.

Reusable, no-code schemas

Marcus defines what “clean” means once per protocol — in dropdowns and chips, not code.

Template name

96-well viability screen v1

Column rule

Remove

Column name

Expected type

Treatment_Group

Categorical



Allowed values

Control x

Drug A x

Drug B x

Drug C x

Type value, press Enter

Why this matters

- Known variants normalize silently so only genuinely new values ever need a human.
- Elena's resolutions can be promoted back into the schema meaning it improves with real usage.
- Detection gets more deterministic and less guess-based for every protocol defined.

What we've already proven



< 3 minutes

raw upload to a downloaded, reproducible script



100%

script-to-cleaned-preview reproducibility



Limited

regex exposure for default-mode users, verified end to end

What's next



Multi-Batch Ingestion

Consolidate weeks of experiments in one upload, not five — automatic reconciliation against one schema.



Universal Instrument Integration

One vendor-agnostic interface for any plate reader, qPCR system, or imaging platform on the bench.



Continuous Audit History

Git-backed versioning of every dataset and protocol change — permanent, queryable institutional memory.

Already built today: protocol/schema authoring layer, and an opt-in Advanced Mode for technical users.

Two different teams. One dataset.

- Pilot this with your lab
- Explore an embedded-partnership integration

Thank you.

Karen Wu · BioBridge OS · QBI Hackathon 2026